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Anisotropy of the emission of DD-fusion neutrons caused by the plasma-focus vessel

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Abstract
The anisotropic emission of DD-fusion neutrons emitted by the plasma focus device PF-1000 at the Institute of Plasma Physics and Laser Microfusion, Warsaw, was observed with the use of thermoluminescent dosimeters and fast-neutron moderators. The observed anisotropy is compared with the computational prediction of the anisotropy caused by the PF-1000 vessel, which scatters the expanding DD-neutrons. The resulting changes in the group energy spectrum of neutrons induced by their scattering on the discharge vessel are presented and their effect on time-of-flight spectra of neutrons is discussed.

1. Introduction

The principle of a dense plasma focus device operation lies in the conversion of energy collected in a capacitor bank into electromagnetic acceleration and compression of short-lived plasma which becomes a source of x-rays, charged particles and nuclear fusion neutrons, when operated, for example, in deuterium [1]. Since the neutron emission lasts tens of nanoseconds [2], the pulsed-beam time-of-flight (TOF) method can be used to advantage [3]. Generally, the TOF spectrum can be analysed to estimate the energy spectrum of neutrons including the broadening of the energy line, which reflects plasma properties [4–6].

The TOF spectra are usually measured outside the plasma vessel. For that reason, the neutron field inside the plasma vessel room has two components: primary fusion neutrons and those colliding with components of the plasma vessel and with room walls. The importance of neutron scattering from laboratory equipment has already been reported [7]. To the best

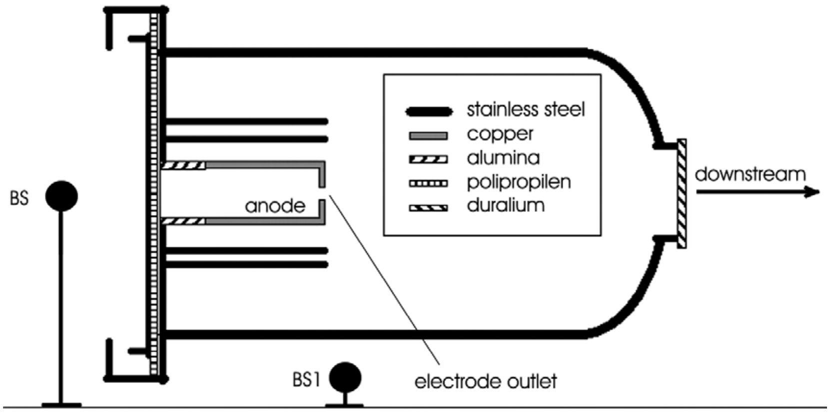

Figure 1. Simplified structure of the plasma-focus vessel for computing the neutron scattering. The vessel of stainless steel is 3.2 m in length; the vacuum chamber is 2.5 m in length and its diameter is 1.4 m; the average wall thickness is 10 mm. 288 cables arriving at the collector ring of 2.5 m in diameter (left) are not sketched. BS1—position of a reference BS containing a pair of dosimeters ^{6}LiF : Mg,Cu,P (TLD-600H) and ^{7}LiF : Mg,Cu,P (TLD-700H). Other BSs positioned around the discharge vessel.

of our knowledge, the effect of plasma vessel walls on the resulting neutron spectrum has not been measured for a plasma-focus vessel.

This paper is devoted to the anisotropy of DD-neutron emission from the plasma focus device PF-1000 at the Institute of Plasma Physics and Laser Microfusion, Warsaw. The anisotropy is measured employing time-integrating detectors composed of Bonner sphere (BS) neutron moderators and thermoluminescent dosimeters (TLDs). The contribution of the plasma-focus vessel on the neutron scattering to the anisotropy was computed using the MCNP 4C code [8]. Except the anisotropy caused by the discharge vessel group spectra of scattered and direct neutrons were determined. This is a unique piece of information, which can elucidate the shape of the observed TOF spectra, because it is not obtainable from any other diagnostic.

2. Experimental arrangement

The pinching plasma was generated in the PF-1000 vessel discharging energy of 450–500 kJ in the deuterium with a pressure of 3.5 Torr. The discharge was created within a large Mather-type electrode system consisting of a copper anode (231 mm in diameter and 600 mm in length) surrounded by 12 outer stainless-steel electrodes 80 mm in diameter, which were positioned at a radius of 200 mm [9]. The pinching plasma reaches a maximum yield of $\approx 3.5 \times 10^{11}$ neutrons per shot [6].

The detected DD-neutrons were moderated in paraffin BSs having a diameter of 10 in. to be detectable by a pair of ^{6}LiF : Mg,Cu,P (TLD-600H) and ^{7}LiF : Mg,Cu,P (TLD-700H) chips [10]. The paraffin sphere serving as a reference detector (BS1) was placed 1 m below the electrode outlet, as figure 1 schematically shows. The other 3 BSs were positioned around the PF-1000 vessel to measure the emission anisotropy in more directions. The TLDs were exposed to irradiation from about 7 shots to obtain a sufficiently high thermoluminescent (TL) signal. Since the BSs were positioned at various distances, the TL signals were recalculated to the solid angle corresponding to the reference BS1. The dimensions of the TLD chips were 3.2 mm × 3.2 mm × 0.9 mm. The lower sensitivity limit of the

LiF : Mg,Cu,P is 1 μGy. TL signals were read out at a heating ramp 3 °C s^{-1} from 50 to 240 °C followed by a reader annealing at 240 °C for 120 s in an N_2 atmosphere using a PC-aided Harshaw Model 3500 reader. The response of a pair of the dosimeters mentioned above to a mixed field of thermal neutrons and x-ray radiation is determined by a relation $R^{n}_{\mathrm{TLD600}} = R^{n+\gamma}_{\mathrm{TLD600}} - k\,R^{n+\gamma}_{\mathrm{TLD700}}$, where $R^{n+\gamma}_{\mathrm{TLD600}}$ is the response of a TLD-600 chip to the mixed field and $R^{n+\gamma}_{\mathrm{TLD700}} = R^{n}_{\mathrm{TLD700}} + R^{\gamma}_{\mathrm{TLD700}} \approx R^{\gamma}_{\mathrm{TLD700}} = R^{\gamma}_{\mathrm{TLD600}}\,k^{-1}$ is the response of a TLD-700 chip to this field. The parameter k equalizing TLD-600H and TLD-700H responses to x-rays was measured at ^{137}Cs photon beam. The number of emitted neutrons per shot was simultaneously measured with four silver activation detectors [6].

TOF spectra of neutrons and hard x-rays filtered by a 75 mm lead were observed with probes composed of fast photo-multipliers and BC408 scintillators. The TOF probes were positioned at distances of 7, 16.3 and 58.3 m from the electrode outlet in the downstream direction, i.e. along the electrode symmetry axis [6].

3. Direct and scattered neutrons

The precise interpretation of detector responses to a neutron field affected by a fusion vessel requires distinguishing the scattered and direct neutrons. The very effective tool of the determination of a fluence of scattered and direct neutrons in a chosen laboratory place as well as of corresponding group energy spectra is evidently the MCNP 4C code [8], which makes the computation possible taking the structure of the discharge vessel and of the surrounding laboratory room into account.

To provide a preliminary analysis of the effect of the plasma-focus vessel on the neutron scattering with the use of a MCNP 4C code, the structure of the discharge vessel was simplified, as figure 1 shows. Therefore the walls of the laboratory as well as 288 high-voltage cables arriving at the collector ring were neither sketched nor taken into account. For the basic elucidation of the neutron scattering effect on the emission anisotropy and on the energy spectrum broadening, only 2.45 MeV mono-energetic deuterium–deuterium neutrons were considered. The computed angular dependence of the fluence of scattered and direct DD-neutrons at a distance of 10 m from the electrode outlet is shown in figure 2. As direct neutrons are considered those that passed through the discharge vessel and surrounding air atmosphere without any collision and, thus, were kept at the original fusion energy of 2.45 MeV. Figure 3 shows the appropriate anisotropy defined as the ratio of total neutron yields $Y_n(\psi)/Y_n(90°)$ allowing for both the direct and scattered neutrons [1], where ψ is an angle of observation related to the downstream direction. The neutron emission anisotropy caused by the PF-1000 vessel ranges from 0.30 to 1.1. It is evident that such a calculation is of tremendous importance for any experiment devoted to determining the anisotropy of neutron emission from a plasma source.

The scattering of neutrons going trough the discharge vessel also results in their slowing down. The group energy distribution of scattered neutrons ranges from about 15 eV up to close to 2.45 MeV and it is angularly dependent, as figure 4 shows for the picked directions $\psi = 0°$ (downstream) and 90°. The computation led to the estimate that at least 54% of the produced neutrons are scattered by the PF-1000 vessel and by the laboratory equipment, etc.

4. Observed neutron emission

4.1. Time-integrated measurement

The earlier experiments using BSs and TLD dosimeters indicated the possibility of formation of the emission anisotropy due to the neutron scattering by the discharge vessel [10]. Figure 5

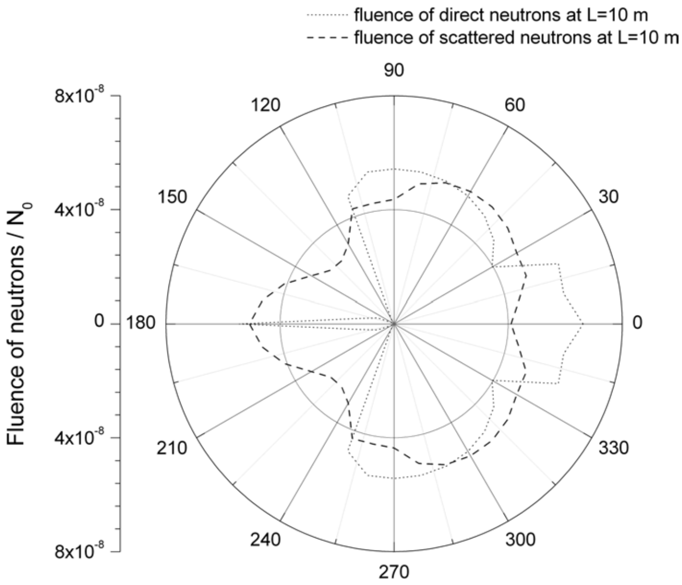

Figure 2. Calculated fluence of scattered and direct neutrons at a distance of 10 m far from the electrode outlet, see figure 1.

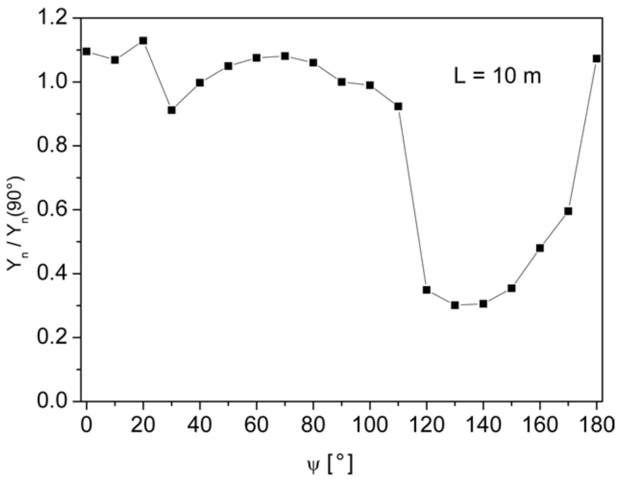

Figure 3. Calculated anisotropy of the neutron emission caused by the PF-1000 vessel to the originally isotropic emission.

shows a comparison of the measured anisotropy, which was determined as the ratio of BS signals to the reference one (in figure 1 labelled as BS1), with the anisotropy computed under the assumption of isotropic emission of DD-neutrons. The computed anisotropy includes direct as well as scattered neutrons and it is related to the angle $\psi = 270°$. Although only a simplified structure of the PF-1000 vessel was considered, the computed anisotropy well elucidates the observed anisotropy mainly for ψ ranging from 25° to 335°. The highest anisotropy, which was observed as well as computed for angles ranging from 110° to 250°, is mainly created by the electrode system. The computed peak at $\psi \approx 180°$, i.e. in the upstream direction, arises along

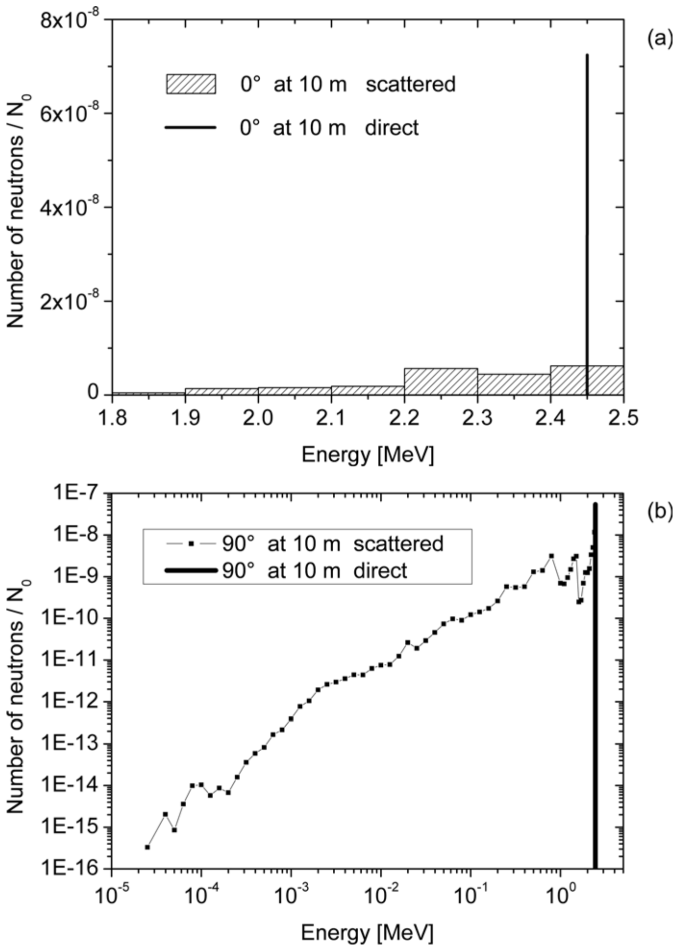

Figure 4. Computed group spectra of scattered and direct DD-neutrons at a distance of 10 m from the electrode outlet at a 0° angle, i.e. the downstream direction, and for neutron energies higher than 1.8 MeV (*a*) and at a 90° angle (*b*).

the axis of the set of massive electrodes where the direct neutrons are passing through a thin vessel wall. The minima at $\psi \approx 30^\circ$ and 330° correspond to the increased effective thickness of the vessel's tube, which is hit by neutrons under the increasing impact angle. The maximum at $\psi \approx 15^\circ$ (345°) was already ascribed to plasma properties because the computing does not demonstrate it [10]. Nevertheless, its origin is not clear. Although the complete structure of the discharge vessel was not taken into account and thus the computed anisotropy could be different from the correct value, the computation renders the basic information on the anisotropy caused by the discharge vessel.

Broadband energy spectra are generally observed with the use of a set of BSs having various diameters. Although no standard BS spectrometer was employed in this experiment, the epithermal neutrons were detected placing the TLDs in a PMMA cylinder (∅1" × 0.85") and in a PE cube ($6 \times 6 \times 6\,\mathrm{cm}^3$) [10]. The appropriate TLD responses correlate with the computed group energy spectra. We should also note that a net TLD-600H response to neutrons was up to about 5 times higher than a response to γ-rays, when the neutron yield reached a value of 1×10^{11}.

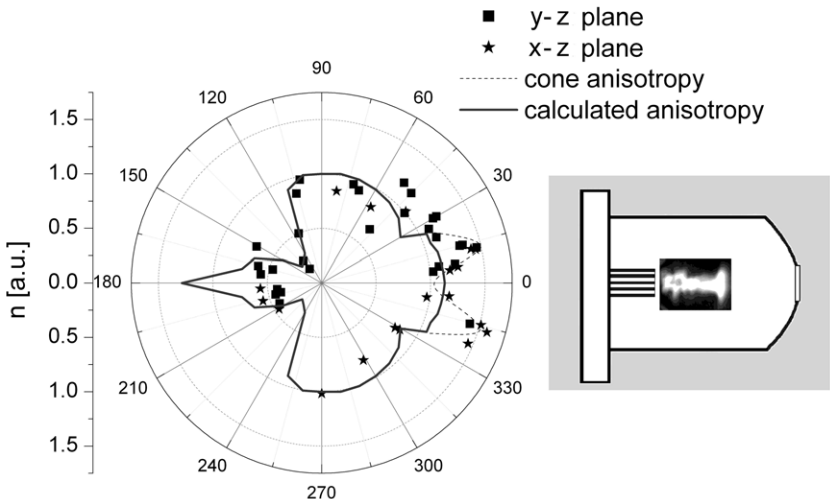

Figure 5. Anisotropy of the neutron emission from the PF-1000 device related to the angle $\psi = 270°$: solid line - anisotropy caused by the structure of the PF-1000 vessel computed with the use of the MCNP 4C code (the simplified structure of the vessel shown in figure 1 was only considered); symbols - anisotropy of DD-neutrons in *XZ*- and *YZ*-planes (horizontal and vertical planes, respectively) measured with pairs of TLD-700H and TLD-600H; dashed line—cone shape anisotropy along the axis symmetry. The PF-1000 facility was operated at an energy of 450–500 kJ. The D_2 pressure was 3.5 Torr.

4.2. TOF measurement

The computed group energy spectra of neutrons emitted by the PF-1000 vessel shown in figure 4 imply that the scattered neutrons should be distinguishable in TOF spectra: the slower scattered neutrons must hit a detector with a delay in comparison with the direct neutrons being the fastest ones and creating the leading edge of TOF spectra. Therefore, it is very important to separate scattered neutrons from direct ones in the TOF spectrum before the transformation of the TOF spectrum in a velocity distribution and consequently in an energy distribution. Without their separation the direct transformation in the velocity or energy distribution is disputable.

Generally, the TOF spectrum is related to the velocity distribution. Since the motion of expanding particles occurs in a 3-dimensional space, the differential $\mathrm{d}\eta$ of a neutron flux in an x-direction is $\mathrm{d}\vec{v} \times v_x f(\vec{v})$. The procedure for deducing the expression for the TOF signal $S(L,t)$ induced by a flux of species with the velocity distribution function $f(\vec{v})$ was done by Kelly and co-workers [11, 12]. It is based on the substitution $v_x = L/t$, $|\mathrm{d}v_x| = L\,\mathrm{d}t/t^2$, $\mathrm{d}v_y = \mathrm{d}y/t$, $\mathrm{d}v_z = \mathrm{d}z/t$; $\mathrm{d}z\,\mathrm{d}y = \mathrm{d}S$ (the detector size) in $\mathrm{d}\eta$:

$$S(L,t) \propto v_x f(\vec{v})\,\mathrm{d}\vec{v} \propto \frac{L^2}{t^5} f\left(\frac{L}{t}\right), \tag{1}$$

where L is the distance from the source and t is the corresponding TOF. Consequently, the velocity distribution can be determined by the fast-detector's signal transformation as

$$f(v) \propto v^{-5} S(L/v), \tag{2}$$

where v is the neutron velocity oriented along the direction from the electrode outlet to the detector.

Although Kelly's procedure was related to the ablation by laser irradiation (i.e. vaporization, sputtering and desorption of atoms and their ionization), it can be applied to other

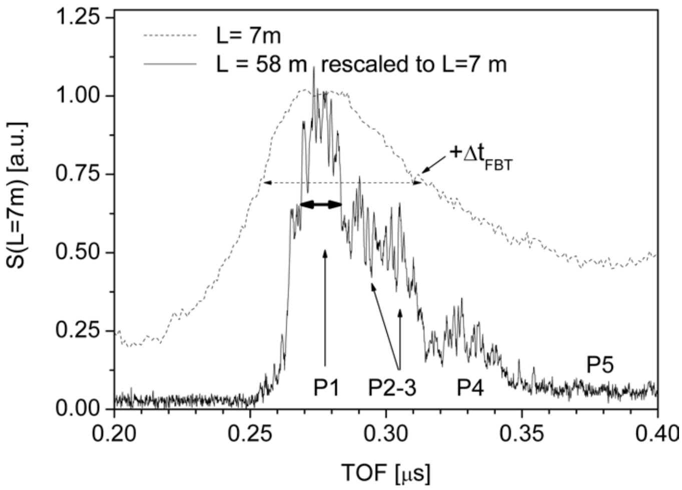

Figure 6. Comparison of TOF signals observed at distances of 7 and 58 m far from the electrode outlet. The time scale of the signal observed at the 58 m distance (solid line) was rescaled to the 7 m distance. The label $+\Delta t_{\mathrm{FBT}}$ illustrates the effect of the fusion burn time on the width of a TOF spectrum if the TOF is not much longer than the fusion burn time. The arrow showing the full width of the P1 peak is at 0.6—maximum due to the attendance of the P2-3 peak.

phenomena where the collisions amongst the particles lead to a velocity distribution function which is commonly approximated as Maxwellian in a centre-of-mass coordinate system. Thus, the formula (1) can join together two characteristics of the emitted DD-neutrons: (a) collisional processes at the beginning of the free neutron expansion, where the velocity distribution of deuterium ions imposes the time profile of the expanding outburst of neutrons due to the temperature; (b) the centre-of-mass velocity of the expanding neutrons, where the velocity of $2.17 \times 10^7\,\mathrm{m\,s^{-1}}$ of 2.45 MeV neutrons is superimposed on the centre-of-mass velocity of the fast deuterium ions.

Equation (1) also reflects the rarefaction phenomenon of expanding species: the increasing distance L results in a decrease in the flux $S(L, t) \sim L^{-3}$ and in an increase in the time width, for example FWHM, $\Delta t_u = L/\Delta u$ of the pulsed flux due to the TOF separation, where Δu is the width of the velocity distribution. If the fusion burn time Δt_{FBT} is comparable to Δt_u, then the total width, Δt_{TOF}, can be expressed as

$$\Delta t_{\mathrm{TOF}} = \Delta t_{\mathrm{FBT}} + L/\Delta u. \tag{3}$$

This formula makes the estimation of both the fusion burn time and the velocity width possible if TOF spectra are observed in the same direction at different distances, as figure 6 shows. The time scale of $S(58\,\mathrm{m},t)$ was rescaled as $t \times 7/58$ and the amplitudes were normalized. The rescaled signal $S(58\,\mathrm{m},t)$ evidently has a shorter time width then $S(7\,\mathrm{m},t)$. Using equation (3), the values of Δt_{FBT} and Δu were estimated to be 45 ns and $8.2 \times 10^8\,\mathrm{m\,s^{-1}}$, respectively. Figure 6 also shows that a short observation distance L results in a broad smudge single peak TOF spectrum with hidden details.

Since the TOF of 2300 ns corresponding to the $S(58\,\mathrm{m},t)$ peak is much longer then the estimated value of the fusion burn time of 45 ns, the $S(7\,\mathrm{m},t)$ signal can hardly show a multi-peak structure as the $S(58\,\mathrm{m},t)$ signal does. As figure 6 demonstrates, the TOF spectrum observed far from the source has a few dominant peaks labelled P1, P2-3 and P4. The slowest peak P5 has a broad width and low amplitude. The side-band peaks P2-3 and P5 could be interpreted as peaks of scattered neutrons interfering with the main pulses of direct neutrons,

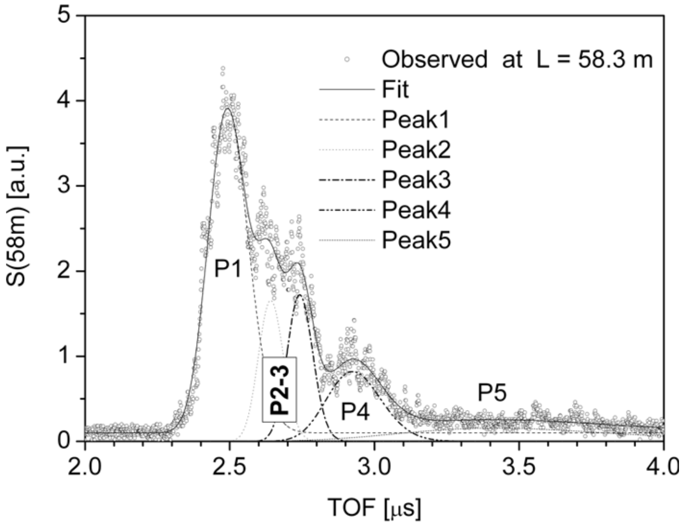

Figure 7. Deconvolution of the TOF spectrum observed at the distance of 58.3 m from the electrode outlet in the downstream direction ($\psi = 0°$).

P1 and P4, generated in two distinct phases of neutron production mechanisms. It is also evident that a half-width of the P1 and P4 peaks cannot be estimated without a spectral analysis of the observed TOF spectrum.

As mentioned earlier, the direct transformation of an observed TOF spectrum in a velocity distribution with the use of equation (2) is disputable due to the significant number of scattered neutrons. The exact solution could be obtained by a numerical simulation of the neutron emission taking at least the neutron scattering by the plasma-focus vessel into account. Nevertheless, a simple partial picture of the neutron scattering effect on the TOF spectrum could be obtained analysing TOF spectra by the use of a general velocity distribution, which can be represented by a shifted Maxwell–Boltzmann distribution. The appropriate TOF spectrum takes the form

$$S(L,t) \propto \frac{L^2}{t^5}\,\exp\left[-\frac{m}{2kT}\left(\frac{L}{t} - u_{\mathrm{CM}}\right)^2\right], \tag{4}$$

where m is the mass of neutron, k is the Boltzmann constant, T is the temperature, u_{CM} is the centre-of-mass velocity of the expanding bunch of neutrons.

Since the z-pinching followed by the fusion burning can appear in the discharge two or more times [1, 13, 14] different instants of time $t_{i,0}$ corresponding to the creation of the z-pinches have to be taken into account. Then the observed detector signal is

$$S(L,t) = \sum_i S_i(L_i, t - t_{i,0}), \tag{5}$$

where S_i is the partial detector's signal induced by neutrons created by the ith z-pinch. Most of the TOF spectra observed at the PF-1000 experiments exhibit a multi-peak structure, as the TOF spectrum $S(58\,\mathrm{m},t)$ shows in figure 7. This figure demonstrates that the $S(58\,\mathrm{m},t)$ spectrum is composed of at least 5 peaks, which were recovered applying expressions (4) and (5) keeping (except the distance of observation L) all 5×4 parameters free, i.e. the amplitude a_i, the temperature T_i, the instant of time $t_{i,0}$ and the centre-of-mass velocity $u_{i,\mathrm{CM}}$. The final interpretation of deconvolution is partially based on the supposed occurrence of scattered neutrons and on a simultaneous observation of two x-ray flashes accompanying the neutron emission, which confirm that the fusion burning was arisen at two different instants [6]. Therefore, considering these actual experimental observations, we will interpret the recovered peaks using values of their parameters fitted to the TOF spectrum as follows:

- Peaks P1 and P4 represent pulsed fluxes of direct neutrons generated by two fusion burns at $t_{1,0} = 0$ ns and $t_{2,0} = 293$ ns, respectively. The corresponding temperatures are $T_1 = 4.3$ keV and $T_4 = 6.5$ keV. The fitted value of $E_{CM} = 2.82$ MeV of the strong peak P1 shows an up Doppler shift $\Delta E_{pl} \approx 370$ keV by the z-pinching plasma, which moves in the downstream direction, while the peak P4 shows only a small shift $\Delta E_{pl} \approx 70$ keV.
- The peaks P2-3 and P5 substitute an undefined behaviour of the detector's response to neutrons slowed down in the discharge vessel body and in the air atmosphere. In this case the values of the fitted parameters do not represent any physical properties of the observed streams but only the shape of the corresponding part of the detector's signal. While the peaks P2-3 reflect the scattered neutrons generated by the first fusion burn, the last peak P5 could be interpreted as a group of scattered neutrons created by both the fusion burns due to the its broad FWHM. The abundance of the scattered neutrons represented by the total area of all three peaks P2, P3 and P5 is about 30%, which is not inconsistent with the computed group spectrum.

Although due to the loss of identity of the space-time origin of scattered neutrons the peaks of scattered neutrons arisen in a TOF signal should not obey the signal function (4), it seems that is just the shape of the function (4), which could simulate a pulsed stream of species being of different origin. Yap *et al* [14] also applied a Maxwellian-pulse fitting technique (but not specified) for resolving the two-phase neutron emission history, which gave the corresponding fusion burn times. The typical fast increase and subsequent gradual decrease in the observed pulses of neutrons could be tolerable simulated by a pulsed curve (4) but without any demand on a physical justification of all its fitted parameters. Nevertheless, it can make the interpretation of TOF spectra possible.

5. Conclusions

Observations of the effect of the plasma-focus vessel structure on the anisotropy of neutron emission were presented. The experimental determination of the anisotropy coefficient defined as the ratio of neutron yields measured in the main directions $Y_n(0°)/Y_n(90°)$ of discharge co-ordinates must be completed by the calculated anisotropy which is caused by the discharge vessel employed. The influence of this anisotropy on time-resolved signals of active neutron detectors has to be taken into account for the correct interpretation and evaluation of the observed TOF spectra needed for the description of the plasma-focus dynamics.

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References

[1] Vikhrev V V and Korolev V D 2007 *Plasma Phys. Rep.* **33** 356–80
[2] Kubeš P *et al* 2006 *Czech. J. Phys. Suppl.* B **56** 274–9
[3] Brooks F D and Klein H 2002 *Nucl. Instrum. Methods* A **476** 1–11

[4] Lerche R A, Coleman L W, Houghton J W, Speck D R and Storm E K 1977 *Appl. Phys. Lett.* **31** 645–7
[5] Murphy T J, Chrien R E and Klare K A 1997 *Rev. Sci. Instrum.* **68** 610–3
[6] Scholz M *et al* 2006 *Czech. J. Phys. Suppl.* B **56** 243–9
[7] Vega-Carrillo H R 2001 *Nucl. Instrum. Methods* A **463** 375–86
[8] Briesmeister J F (ed) 2000 MCNP^{TM}– A General Monte Carlo N-Particle Transport Code, Version 4C *Los Alamos National Laboratory Report* LA-13709-M
[9] Scholz M *et al* 2006 *Nukleonika* **51** 79–84
[10] Velyhan A *et al* 2006 *Phys. Scr.* T **123** 112–5
[11] Kelly R and Dreyfus R W 1988 *Surf. Sci.* **198** 263
[12] Miotello A and Kelly R 1999 *Appl. Surf. Sci* **138–139** 44
[13] Scholz M *et al* 2004 *Vacuum* **76** 361–4
[14] Yap S L, Wong C S, Choi P, Dumitrescu C and Moo S P 2005 *Japan. J. Appl. Phys.* **44** 8125–32